Karthik M

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Education

Amrita Vishwa Vidyapeetham, Chennai 2022 – 2026
B.Tech, Automation and Robotics Engineering; Minor in AI and Machine Learning *CGPA 9.26 / 10*
- Rank 1 in the department and Gold Medalist; Top Performer Award for consecutive semesters.

Experience

Perficient Sep 2026 – Present
Associate Technical Consultant, Product Development (Global Graduate Training Program) *India*
- Build full-stack products end to end in React 19, Spring Boot 3 (Java 21), PostgreSQL and FastAPI, planned and delivered through GitHub milestones, issues and pull requests.
- Delivered a wealth-management platform joining digital account opening with an advisor workspace across five roles, backed by 273 automated tests (JUnit, pytest, Vitest, Playwright).
- Deployed a hospital management system (Spring Boot, FastAPI, React) to Google Kubernetes Engine, and an e-commerce application to a Compute Engine VM with Docker Compose.
- Found and fixed defects in Perficient's AI Platform Context and Documents Handler as part of a team of five, each fix proven with before-and-after tests in a reviewed pull request.

Caterpillar Inc. Jan 2026 – Jul 2026
Data Scientist Intern, Machine Telemetry Analytics Platform *Bengaluru*
- Built real-time data pipelines over high-volume machine telemetry for a 16-member platform team.
- Engineered Snowflake data models and traced data-quality issues back to their source systems.
- Built the dashboards engineering teams used to track machine performance, and automated recurring reporting with Playwright and python-pptx.
- Presented the Grouped Query Attention paper in the team's research reading series.

Amazon May 2025 – Jun 2025
Data Engineer Intern, CMT VarOps *Bengaluru*
- Built a Text-to-SQL assistant on Amazon Bedrock with a retrieval-augmented generation pipeline, deployed on AWS Lambda and used across a 162-member division.
- Designed agent guardrails and multi-agent task decomposition to keep generated SQL accurate and safe to run.

Projects

Wealth NAO + Advisor 360 | *Spring Boot 3.5, Java 21, FastAPI, React 19, PostgreSQL, scikit-learn* GitHub: coming soon
- Built one platform for a wealth firm, from a 13-step client application (identity and anti-money-laundering checks, risk profile, FATCA/CRS tax declarations) to an advisor's Client 360, household and risk views.
- Secured both services with RS256-signed JWTs verified through a shared JWKS endpoint; encrypted sensitive fields and documents with AES-256-GCM; enforced four-eyes compliance approval and a tamper-evident audit trail.
- Added a Python insights service: Holt forecasting of assets under management and a logistic-regression attrition model (AUC 0.773), shown in interactive Chart.js treemap, Sankey and heatmap views.

ROSRAG: Retrieval-Augmented Assistant for ROS2 | *Python, FAISS, SentenceTransformers, Gemini, Streamlit* PyPI
- Published to PyPI: FAISS search over ROS2 documentation with MiniLM embeddings and grounded answers from Gemini 2.5 Flash, with a CLI and a Streamlit interface.
- Built an evaluation harness before trusting the model, reaching Precision@5 of 0.91 and Mean Reciprocal Rank of 0.94.

Autonomous UAV with Robotic Arm | *C++, ROS2, Gazebo* SAEINDIA: 2nd runner-up
- Owned perception, control and integration for a drone that navigates, picks and places a payload with no pilot: Extended Kalman Filter sensor fusion at 200 Hz and inverse-kinematics arm control.
- Reached ±2 cm placement accuracy, verified in simulation and field tests; placed 3rd of 74 teams.

statelet: Multi-App Onboarding Agent | *Python, Slack, Notion, Linear, Google Calendar APIs* GitHub
- Built in one day a declarative reconciler that provisions a new hire across four apps, verifying every write and rolling back on failure.

GPU-Accelerated Compute Engine | *C++, CUDA, Nsight Compute*
- Wrote a CUDA dense linear-algebra library (matrix multiply, LU decomposition, iterative solvers) using shared-memory tiling and warp-level primitives; up to 18× faster than CPU, benchmarked against cuBLAS.

End-to-End Data Platform | *AWS Lambda, Airflow, PySpark, Apache Iceberg, S3, Snowflake*

- Built event-driven ingestion with Lambda, Airflow orchestration and Spark transforms into partitioned Iceberg tables with ACID guarantees across raw, curated and serving zones.

Retail Shelf Analytics and Auto-Billing | *PyTorch, Faster R-CNN, YOLOv11, OpenCV*
- Compared two-stage and single-stage detectors for product recognition on cluttered shelves, and improved occluded cases with targeted augmentation.

Ball-Balance Control with Soft Actor-Critic | *MATLAB, Simulink, Simscape*
- Designed actor-critic networks and reward shaping to stabilise a manipulator-mounted plate over a 22-dimensional state space on a physics-accurate 6-DOF model.

Statistical Arbitrage Backtester | *Python, pandas, statsmodels*
- Built pairs-trading research with cointegration tests and Z-score mean-reversion signals, measuring risk with Monte Carlo simulation and Value-at-Risk.

Technical Skills

Languages: Python, C++, C, Java, JavaScript, TypeScript, SQL, MATLAB, Bash, HTML, CSS

Full-stack and APIs: React 19, Redux Toolkit, Vite, Tailwind CSS, Chart.js, Node.js, Spring Boot, Spring Security, Spring Data JPA, Hibernate, FastAPI, Pydantic, REST, OpenAPI/Swagger, JWT, Microservices, Streamlit

AI and Machine Learning: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, LLM agents, multi-agent systems, retrieval-augmented generation, FAISS, Amazon Bedrock, Gemini, Strands Agents SDK, computer vision (YOLO, Faster R-CNN, OpenCV), NLP, reinforcement learning, time-series forecasting

Robotics and Embedded: ROS2/DDS, Gazebo, Extended Kalman Filter, inverse kinematics, control systems, Simulink, Simscape, ESP32, ARM, Industrial IoT, edge machine learning

Data Engineering: Apache Spark, Airflow, Snowflake, Apache Iceberg, PostgreSQL, Flyway, ETL/ELT, data modelling, data quality, dashboards

Cloud and DevOps: AWS (Lambda, S3, Bedrock), Google Cloud (Compute Engine, GKE, Cloud Storage), Docker, Docker Compose, Kubernetes, nginx, CI/CD, Git, GitHub, Linux

Testing and Security: JUnit 5, Mockito, pytest, Vitest, React Testing Library, Playwright, JaCoCo, test-driven development; AES-256-GCM, BCrypt, two-factor sign-in, role-based access control

High-Performance Computing and Quant: CUDA, Nsight Compute, profiling; probability, statistics, Monte Carlo simulation, Value-at-Risk

Publications and Talks

- **LoDISA 2026, University of Cambridge** (accepted and presented): IIoT air-quality monitoring with edge sensing and machine-learning analytics for high-dust environments, validated on the UCI Air Quality dataset with a Random Forest model. With Swamisudhan M, Bhavesh Durai S, Jaswanth G, Charan Aravvind S and Dr Rishi Kumar.
- **NeurIPS 2026 Creative AI Track** (submitted): *Marks Against the Wind*, a drone and robotic-arm system treated as a creative agent.
- **Invited Speaker, UbuCon India 2025, IISc Bangalore**: embedded robotics and autonomous ROS2 systems.

Achievements and Leadership

- Ranked 46 of 34,000+ teams, Amazon ML Challenge; macro F0.5 of 0.958 in the 2026 Business Entity Resolution challenge.
- 2nd runner-up (3rd of 74), SAEINDIA Autonomous Drone Challenge, leading the 10-member DART team; the only team from Tamil Nadu recognised.
- National Finalist, ICSC 2025; Silver Honour, International Youth Math Challenge 2023; CodeChef 4-star with 500+ problems solved.
- Hackathons: Multi-App AI Agent Hackathon, AWS Agents for Humans, Razorpay AI Buildathon (Track 03, AI Revenue Recovery).
- Founder and Chair, IEEE Robotics and Automation Society student chapter, Amrita.